

Hosting a Simple HTML Web Application on Apache

To Begin with the Lab

Summary of the Lab

This lab demonstrated hosting a simple HTML website using the Apache web server on Ubuntu. It covered updating packages, installing and verifying Apache, accessing the web root directory, modifying permissions, replacing the default webpage with a custom index.html, verifying the changes in a browser, and stopping the Apache service after completion.

- Update the Package Index

```
sudo apt update
```

- Install Apache

```
sudo apt install apache2 -y
```

- Verify Apache
- Open your browser:
- `http://<EC2-Public-IP>`
- You should see the default Apache2 Ubuntu Default Page.



- Navigate to Apache's Web Root

```
cd /var/www/html
```

- List the files:

```
ls
```

- You should see:
- index.html

```
ubuntu@ip-172-31-45-84:~$ cd /var/www/html
ubuntu@ip-172-31-45-84:/var/www/html$ ls
index.html
ubuntu@ip-172-31-45-84:/var/www/html$
```

- Change Directory Permissions

- Give write permissions to the web root directory.

```
sudo chmod -R 777 /var/www/html
```

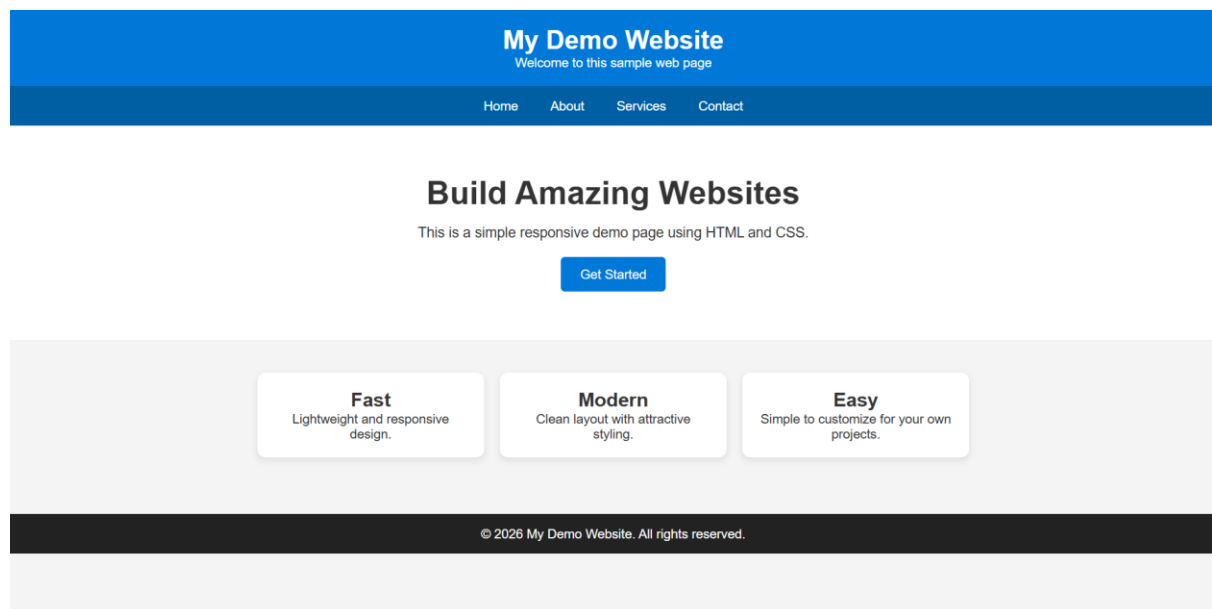
- Remove the Default Web Page
- `sudo rm /var/www/html/index.html`
- Verify:
- `ls /var/www/html`
- The default `index.html` should no longer exist.

```
ubuntu@ip-172-31-45-84:/var/www/html$ sudo chmod -R 777 /var/www/html
ubuntu@ip-172-31-45-84:/var/www/html$ sudo rm /var/www/html/index.html
ubuntu@ip-172-31-45-84:/var/www/html$ ls /var/www/html
ubuntu@ip-172-31-45-84:/var/www/html$
```

- Now create a new `index.html` in the same directory `/var/www/html`

```
ubuntu@ip-172-31-45-84:/var/www/html$ sudo chmod -R 777 /var/www/html
ubuntu@ip-172-31-45-84:/var/www/html$ sudo rm /var/www/html/index.html
ubuntu@ip-172-31-45-84:/var/www/html$ ls /var/www/html
ubuntu@ip-172-31-45-84:/var/www/html$ nano index.html
ubuntu@ip-172-31-45-84:/var/www/html$ ls
index.html
ubuntu@ip-172-31-45-84:/var/www/html$ pwd
/var/www/html
ubuntu@ip-172-31-45-84:/var/www/html$
```

- And refresh the browser and the Apache webpage change to your webpage



- When you're finished and ready for the Docker lab:
- `sudo systemctl stop apache2`
- Verify it has stopped:
- `sudo systemctl status apache2`

```
ubuntu@ip-172-31-45-84:/var/www/html$ sudo systemctl stop apache2
ubuntu@ip-172-31-45-84:/var/www/html$ sudo systemctl status apache2
o apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: inactive (dead) since Wed 2026-07-08 05:45:41 UTC; 51s ago
     Duration: 56min 22.616s
  Invocation: 3010c22f98ae47baae1fafbebe67eed6
     Docs: https://httpd.apache.org/docs/2.4/
    Process: 5081 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
    Process: 6026 ExecStop=/usr/sbin/apachectl graceful-stop (code=exited, status=0/SUCCESS)
   Main PID: 5081 (code=exited, status=0/SUCCESS)
    Status: "Total requests: 51; Idle/Busy workers 100/0;Requests/sec: 0.0151; Bytes served/sec: 15 B/sec"
   Mem peak: 9.4M
      CPU: 366ms

Jul 08 04:49:19 ip-172-31-45-84 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Jul 08 04:49:19 ip-172-31-45-84 systemd[1]: Started apache2.service - The Apache HTTP Server.
Jul 08 05:45:41 ip-172-31-45-84 systemd[1]: Stopping apache2.service - The Apache HTTP Server...
Jul 08 05:45:41 ip-172-31-45-84 systemd[1]: apache2.service: Deactivated successfully.
Jul 08 05:45:41 ip-172-31-45-84 systemd[1]: Stopped apache2.service - The Apache HTTP Server.
ubuntu@ip-172-31-45-84:/var/www/html$
```

Build a Custom Docker Image

To Begin with the Lab

Summary of the Lab

Create a project directory, add a Dockerfile using the httpd image, and build a custom Docker image. Run the container on port 80, verify the application through the EC2 public IP, then stop the container and confirm the website becomes inaccessible.

- Now move to the directory

```
cd /home/ubuntu
```

- create a folder here

```
mkdir app
```

- Go to the directory

```
cd app
```

- Create a file named **Dockerfile**.

```
nano Dockerfile
```

- Add the following content:

```
FROM httpd  
COPY . /usr/local/apache2/htdocs
```

- Create index.html
- Save and exit.

```
ubuntu@ip-172-31-45-84:~$ cd app  
ubuntu@ip-172-31-45-84:~/app$ ls  
Dockerfile  index.html  
ubuntu@ip-172-31-45-84:~/app$
```

- Build the image from the current directory.

```
sudo docker build -t static-app .
```

- docker build → Builds a Docker image.
- -t static-app → Assigns the image name static-app.
- . → Uses the current directory as the build context.

```

ubuntu@ip-172-31-45-84:~/app$ sudo docker build -t static-app .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  5.632kB
Step 1/2 : FROM httpd
latest: Pulling from library/httpd
1eb2ba022f6d: Pulling fs layer
c9b613c6caa5: Pulling fs layer
17ff7bac6cc6: Pulling fs layer
4f4fb700ef54: Pulling fs layer
a31b3e3010ee: Pulling fs layer
e3fff84cea83: Download complete
1eb2ba022f6d: Download complete
c9b613c6caa5: Download complete
17ff7bac6cc6: Download complete
4f4fb700ef54: Download complete
c9b613c6caa5: Pull complete
4f4fb700ef54: Pull complete
a31b3e3010ee: Download complete
e7f6f34d7a39: Download complete
17ff7bac6cc6: Pull complete
1eb2ba022f6d: Pull complete
a31b3e3010ee: Pull complete
Digest: sha256:393435eela31437adeb1f03c134224c9ce5fa5f527e8c0cb9bea576b0d6fc742
Status: Downloaded newer image for httpd:latest
--> 393435eela31
Step 2/2 : COPY . /usr/local/apache2/htdocs
--> 697da8395c47
Successfully built 697da8395c47
Successfully tagged static-app:latest
ubuntu@ip-172-31-45-84:~/app$

```

- List the available Docker images.

```
sudo docker images
```

- You should see:
 - static-app, httpd (downloaded automatically if not already present) and other images present.

```

ubuntu@ip-172-31-45-84:~/app$ sudo docker images

```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
httpd:latest	393435eela31	177MB	47.6MB	
nginx:latest	ec4ed8b5299e	241MB	66MB	🔍
static-app:latest	697da8395c47	175MB	45.2MB	
ubuntu:latest	b7f48194d4d8	160MB	45.3MB	🔍

- List running containers.

```
sudo docker ps
```

- Stop any running containers that may already be using port 80.

```
sudo docker stop <container-id>
```

- No container are running

```

ubuntu@ip-172-31-45-84:~/app$ sudo docker images

```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
httpd:latest	393435eela31	177MB	47.6MB	
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static-app:latest	697da8395c47	175MB	45.2MB	
ubuntu:latest	b7f48194d4d8	160MB	45.3MB	🔍

```

ubuntu@ip-172-31-45-84:~/app$ sudo docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED   STATUS    PORTS   NAMES
ubuntu@ip-172-31-45-84:~/app$

```

- Create and start a container from the custom image.

```
sudo docker run -d -p 80:80 static-app
```

- Docker returns the **Container ID**.

```
ubuntu@ip-172-31-45-84:~/app$ sudo docker run -d -p 80:80 static-app
030e114d0a9bb5c76caf8940fe76e3761b5bfc85b09a6720faad33ea966905f4
ubuntu@ip-172-31-45-84:~/app$
```

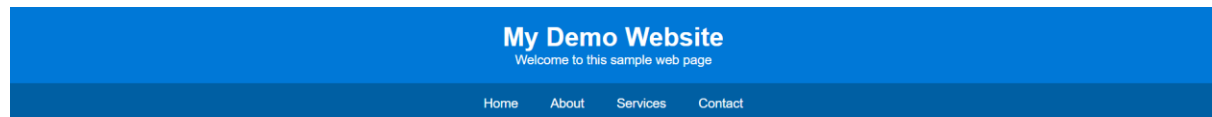
- Verify the Running Container

```
sudo docker ps
```

- Confirm that the static-app container is running.

```
ubuntu@ip-172-31-45-84:~/app$ sudo docker run -d -p 80:80 static-app
030e114d0a9bb5c76caf8940fe76e3761b5bfc85b09a6720faad33ea966905f4
ubuntu@ip-172-31-45-84:~/app$ sudo docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
030e114d0a9b   static-app    "httpd-foreground"      26 seconds ago Up 26 seconds  0.0.0.0:80->80/tcp, [::]:80->80/tcp  nostalgic_nobel
ubuntu@ip-172-31-45-84:~/app$
```

- Open a browser and navigate to:
- <http://<EC2-Public-IP>/index.html>
- Your HTML web application should be displayed.



Build Amazing Websites

This is a simple responsive demo page using HTML and CSS.

[Get Started](#)

- Stop the running container.

```
sudo docker stop <container-id>
```

```
ubuntu@ip-172-31-45-84:~/app$ sudo docker stop 030
030
ubuntu@ip-172-31-45-84:~/app$
```

- Verify the Application Stops
- Refresh the browser.
- The web page will no longer load because the container has been stopped.

